

# ONT METAGENOMICS WORKSHOP

AT [HOST INSTITUTION]

*Metagenomic Next Generation Sequencing  
(Oxford Nanopore/ONT) for Viral  
Identification and Discovery*

[WORKSHOP DATES]

# OVERVIEW OF WORKSHOP

Metagenomic next-generation sequencing (mNGS) is an essential tool for pathogen diagnosis, discovery, and surveillance. This workshop is divided into two core modules:

## Module 1: Wet Lab (Week 1)

Participants will engage in hands-on laboratory sessions to master the fundamentals of sample preparation and genomic library construction. Throughout the week, they will generate and sequence their own mNGS libraries, gaining practical experience with each step.

## Module 2: Dry Lab (Week 2)

The second week transitions into data analysis and interpretation. Using tools such as CZID, participants will explore how to identify potential pathogens from sequencing data and contextualize their results within a broader epidemiological framework.

| Schedule for Week 1 - Wet Lab ([Wet Lab Location]) |                                       |                                                                           |                 |
|----------------------------------------------------|---------------------------------------|---------------------------------------------------------------------------|-----------------|
| Day                                                | Topic                                 | Details                                                                   | Location        |
| [Mon, Date]                                        | Introduction to Genomics              | Basic concepts in genomics, epidemiology, and mNGS                        | Classroom       |
| [Tue, Date]                                        | Theory + Lab Day 1                    | ONT Library Prep Theory, SPRI practice                                    | Classroom + Lab |
| [Wed, Date]                                        | Lab Day 2                             | Sample QC, RNA Library preparation                                        | Classroom + Lab |
| [Thu, Date]                                        | Lab Day 3                             | RNA Library Pooling and QC                                                | Classroom + Lab |
| [Fri, Date]                                        | Lab Day 4                             | Sequencer loading                                                         | Classroom + Lab |
| Schedule for Week 2 - Dry Lab ([Dry Lab Location]) |                                       |                                                                           |                 |
| Day                                                | Topic                                 | Details                                                                   | Location        |
| [Mon, Date]                                        | mNGS Review + Basic Bioinformatics    | Re-introduction of metagenomics theory and introduction to bioinformatics | Classroom       |
| [Tue, Date]                                        | Sequencing data analysis on CZID      | Overview of the platform and looking at run from previous week            | Classroom       |
| [Wed, Date]                                        | Genomic Epidemiology                  | Overview of key concepts in genomic epidemiology, including case studies  | Classroom       |
| [Thu, Date]                                        | Novel virus discovery + Mystery cases | Steps involved in identifying novel viruses from sequencing data          | Classroom       |
| [Fri, Date]                                        | Additional topics + Review            | Final review and conclusion                                               | Classroom       |

# MODULE 1:

## WET LAB

*Hands-on RNA Library Preparation and Sequencing*

**[WEEK 1 DATES]**

# MODULE 1: HANDS-ON RNA LIBRARY PREPARATION AND SEQUENCING

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## Getting Started

- Before starting and after completing your work, always wipe the workspace with RNaseZap/RNaseAway and 70% EtOH
- Tie back long hair
- Wear a clean lab coat
- Change out gloves often
- Organize your workspace in a way that avoids hovering over clean tips/tubes with a used pipette
- Only keep necessary materials out for the protocol you are performing
- Clean the workspace at the end of the day with 70% EtOH and dispose of trash appropriately

For more information on RNAses we recommend visiting:

<https://www.thermofisher.com/us/en/home/references/ambion-tech-support/nuclease-enzymes/tech-notes/how-to-maintain-an-rnase-free-lab.ht>

## List of Protocols

Below is the list of protocols we will be using during this week:

- Protocol 1: SPRI Bead Clean-up
- Protocol 2: Nanopore RNA Library Preparation

# WL- Protocol 1: SPRI Bead Clean-up

## Overview

SPRI (Solid Phase Reverse Immobilization) beads are useful for purifying and size-selecting nucleic acids such as PCR amplicons, NGS libraries, RNA/DNA, and gDNA. Nucleic acids bind to the SPRI beads, are washed, then eluted from the beads before collection. Adjusting the ratio of SPRI beads to sample volume accomplishes selection for nucleic acids by base pair size.

## Protocol

Use either [SPRI Select](#) or [Ampure XP](#) beads. They work equivalently and only differ in storage conditions. This protocol is an adaptation of the manufacturer's.

Based on the desired SPRI ratio and sample volume, calculate the volume of SPRI beads to add. Also decide on a desired final volume after clean-up.

|                              |  |
|------------------------------|--|
| Sample volume:               |  |
| SPRI bead ratio:             |  |
| SPRI to add (per rxn):       |  |
| Elution volume (per rxn):    |  |
| Aspiration volume (per rxn): |  |

1. Add SPRI beads to sample and mix thoroughly by pipetting up and down ten times
2. You may pulse spin the tubes and spin very briefly, but ensure beads are still fully suspended. Incubate for 3-5 mins at room temperature.
3. Place samples on magnetic rack, and incubate for ~5 mins on the rack.
4. Remove supernatant.
5. Add 200µL of 85% EtOH to samples while on the magnetic rack. Do not disturb beads. Incubate at room temperature for 30s then remove the supernatant.
6. Repeat EtOH wash step for a total of 2 washes.
7. Air dry the beads for up to 3-5 mins while on the magnetic rack. *Watch beads carefully throughout this incubation to avoid overdrying; smaller bead pellets dry faster. Trust your eyes more than the timer.*
8. Remove tube from magnetic rack. Add desired elution volume of nuclease free water.
9. Vortex or pipette to mix well. Spin tubes and incubate for a *minimum* of 2 mins at room temperature off the magnetic rack.
10. Place on magnetic rack until solution is clear ~ 5 mins.
11. Remove desired aspiration volume of the supernatant and transfer to a clean nuclease free PCR tube

## **SPRI Bead Clean-up Practice**

We will practice SPRI bead cleaning by performing size selections at different bead:sample ratios on a 50bp or 100bp DNA ladder at a **1:50 dilution**.

1. Create a 1:50 dilution of DNA ladder in a labeled 1.5mL tube.
2. Keep an aliquot of the 1:50 dilution for HS dsDNA Qubit quantification and for a capillary gel electrophoresis run.
3. Prepare sufficient 85% EtOH
4. Perform SPRI clean-up using the protocol in the previous page for the ratios and volumes shown below

| Well             | 1                 | 2                 |
|------------------|-------------------|-------------------|
| Sample           | 20 $\mu$ L ladder | 20 $\mu$ L ladder |
| SPRI Ratio       | 2x                | 0.8x              |
| SPRI bead volume | 40 $\mu$ L        | 16 $\mu$ L        |

### **Check DNA Elution Using dsDNA Qubit**

1. Quantify the initial 1:50 diluted ladder input and all of the SPRI elutions by HS dsDNA Qubit to assess yield.
2. Follow instructions provided by the manufacturer. Each HS dsDNA Qubit assay requires at least 1uL of sample.

Input ladder concentration =

| Sample                      | 1 | 2 |
|-----------------------------|---|---|
| Concentration (ng/ $\mu$ L) |   |   |
| Efficiency (%)              |   |   |

# WL- Protocol 2: Nanopore RNA Library Preparation

## ONT RNA metagenomics workflow

- Sample selection
- RNA extraction
  - (DNase I treatment)
- Input RNA QC
  - Qubit
  - (Tapestation, BioAnalyzer, or QSep)
- Library prep
  1. Annealing
  2. Reverse transcription + template switching
  3. PCR amplification + barcoding
  4. SPRI cleanup
  5. QC of libraries
    - Qubit
    - (Tapestation, BioAnalyzer, QSep, or diagnostic PCR)
  6. Pooling of libraries
    - QC of pool
- Sequencing
  1. Sequencing adapter ligation
  2. MinION loading
- Data analysis
  - fastq concatenation
  - Upload to CZ ID
  - Analysis (see worksheet)
- Confirmatory testing, if warranted

## [WORKSHOP NAME] ONT mNGS workshop

### Sample list

| Workshop ID | Sample category | Sample type | Input RNA concentration (ng/ul) | ONT Barcode | Library concentration (ng/ul) |
|-------------|-----------------|-------------|---------------------------------|-------------|-------------------------------|
| G1_1        | Unknown         |             |                                 |             |                               |
| G1_2_NTC    | NTC             |             |                                 |             |                               |
| G1_3        | Unknown         |             |                                 |             |                               |
| G1_4        | Unknown         |             |                                 |             |                               |
| G1_5        | Unknown         |             |                                 |             |                               |
| G1_6        | Unknown         |             |                                 |             |                               |
| G2_1        | Unknown         |             |                                 |             |                               |
| G2_2_NTC    | NTC             |             |                                 |             |                               |
| G2_3        | Unknown         |             |                                 |             |                               |
| G2_4        | Unknown         |             |                                 |             |                               |
| G2_5        | Unknown         |             |                                 |             |                               |
| G2_6        | Unknown         |             |                                 |             |                               |
| G3_1        | Unknown         |             |                                 |             |                               |
| G3_2_NTC    | NTC             |             |                                 |             |                               |
| G3_3        | Unknown         |             |                                 |             |                               |
| G3_4        | Unknown         |             |                                 |             |                               |
| G3_5        | Unknown         |             |                                 |             |                               |
| G3_6        | Unknown         |             |                                 |             |                               |
| G4_1        | Unknown         |             |                                 |             |                               |
| G4_2_NTC    | NTC             |             |                                 |             |                               |
| G4_3        | Unknown         |             |                                 |             |                               |
| G4_4        | Unknown         |             |                                 |             |                               |
| G4_5        | Unknown         |             |                                 |             |                               |
| G4_6        | Unknown         |             |                                 |             |                               |

### Sample set selection per batch

- Each batch **must** include at minimum 1 negative water control included from extraction through library prep
- It is recommended to include 1 positive control sample that has already tested positive by PCR or sequencing
- It is possible to pool up to 12 samples + 2-4 negative controls per MinION flow cell

## Library Preparation Overview

This protocol is an adaptation of the SMART-9n protocol from protocols.io, with some modifications and updated for the V14 (R10.4) chemistry. The protocol has the following steps:

1. Annealing; 2. Reverse transcription; 3. PCR amplification; 4. SPRI clean-up; 5. Library QC; 5.5. (optional) Secondary Amplification; 6. Pooling + QC; 7. Adapter ligation; 8. Loading

## Protocol

This worksheet is designed to assist with setting up library preparation reactions in the lab. For each step, a master mix can be assembled to streamline the workflow by combining reagents shared across all reactions. Components included in the master mix are highlighted in light pink. The total volume of the master mix is highlighted in dark pink.

For example, to interpret the table for the annealing reaction step below, combine the RLB RT-9N primer and dNTP mix in a clean tube. Mix thoroughly by flicking the tube and briefly spin down. Then, add 2  $\mu\text{L}$  of this master mix to each reaction tube containing 10  $\mu\text{L}$  of RNA sample.

### 1. Annealing

| Reagent                              | 1 rxn                       | X full rxns                             |  |
|--------------------------------------|-----------------------------|-----------------------------------------|--|
|                                      |                             | X =                                     |  |
|                                      | vol stock ( $\mu\text{L}$ ) | vol stock ( $\mu\text{L}$ )             |  |
| RNA sample (10 ng total)             | 10                          | 10 each                                 |  |
| RLB RT-9N primer (20 $\mu\text{M}$ ) | 1                           |                                         |  |
| dNTP mix (10 mM)                     | 1                           |                                         |  |
| Master Mix volume                    | 2                           |                                         |  |
| <b>Total volume</b>                  | <b>12</b>                   | <b>12 each</b>                          |  |
| Pink = Reagents to add to master mix |                             | Add 2 $\mu\text{L}$ of MM to RNA sample |  |

Mix by gently pipetting up and down and incubate in a thermocycler:

· 5 min at 65°C

Snap cool on ice block for 2 min

## 2. Reverse Transcription + Template Switching

| Reagent                              | 1 rxn          | X full rxns                    |  |
|--------------------------------------|----------------|--------------------------------|--|
|                                      |                | X =                            |  |
|                                      | vol stock (μL) | vol stock (μL)                 |  |
| Annealed RNA                         | 12             | 12 each                        |  |
| SSIV buffer (5X)                     | 4              |                                |  |
| RNase OUT (100mM)                    | 1              |                                |  |
| DTT (100 mM)                         | 1              |                                |  |
| SS IV RTase (200 U/μL)               | 1              |                                |  |
| RLB TSO (75 μM)                      | 1              |                                |  |
| Master Mix volume                    | 8              |                                |  |
| <b>Total volume</b>                  | <b>20</b>      | <b>20 each</b>                 |  |
| Pink = Reagents to add to master mix |                | Add 8 μL of MM to annealed RNA |  |

Mix by gently pipetting up and down, then pulse spin.

Incubate reaction mix as follows:

- 90 min at 42°C
- 10 min at 70°C

## 3. PCR Amplification

| Reagent                              | 1 rxn          | X full rxns                                        |   |
|--------------------------------------|----------------|----------------------------------------------------|---|
|                                      |                | X =                                                | 7 |
|                                      | vol stock (μL) | vol stock (μL)                                     |   |
| Tagmented cDNA                       | 5              | 5 each*                                            |   |
| Nuclease-free water                  | 19.5           |                                                    |   |
| RLB (01-24, at 10 μM)**              | 0.5            | 0.5 μL unique to each sample                       |   |
| LongAmp Taq 2x MM                    | 25             |                                                    |   |
| Master Mix volume                    | 44.5           |                                                    |   |
| <b>Total volume</b>                  | <b>50</b>      | <b>50 each</b>                                     |   |
| Pink = Reagents to add to master mix |                | Add 44.5 μL of MM and 0.5 μL RLB to tagmented cDNA |   |

\* Note only 5ul of the cDNA is added to the PCR reaction

\*\*Add barcodes last

Mix MM and final rxn. by pipeting up and down and start the following thermocycling program:

| Thermocycler (lid at 105°C) | Cycles |
|-----------------------------|--------|
| 95 °C for 45s               | 1      |
| 95 °C for 15s               | 26     |
| 59 °C for 15s               |        |
| 65 °C for 5 min             |        |
| 65 °C for 10 mins           | 1      |
| Hold at 10°C                |        |

#### 4. SPRI Bead Clean-up

|                                   |     |    |  |
|-----------------------------------|-----|----|--|
| SPRI bead ratio:                  | 0.6 |    |  |
| SPRI to add (per full rxn):       | 30  | μL |  |
| Elution volume (per full rxn):    | 17  | μL |  |
| Aspiration volume (per full rxn): | 15  | μL |  |

**SPRI clean-up protocol:** Refer to Page 5

**\*\*Checkpoint:** Samples can be stored frozen at -20°C and library prep resumed the next day.

#### 5. QC of individual libraries

Qubit dsDNA HS: ng/ul concentration\* of each sample

Tapestation/Qsep, Genomic DNA: nM concentration\* and size distribution of each sample

Nanodrop dsDNA (optional): identify potential contaminants that may affect pore health

\* If individual libraries are too low concentration at this stage, consider moving to step 5.5. Secondary Amplification (Appendix)

#### 6. Pooling

Pool all barcoded libraries in the desired ratios to a total of 100-150 fmol in 11 μL of water (~10-15nM) or elution buffer (EB). Key points to remember when pooling:

- Aim to have roughly equal amounts of each sample
- For water controls - use 1/5<sup>th</sup> or 1/10<sup>th</sup> the amount
- Don't use more than half the volume of the sample in the pool
- To determine molar amount, use concentration from Qubit and size estimate
- See Library Pooling slides and Pooling Calculation Worksheet for more information
- NEB calculator for estimating molarity: <https://nebiocalculator.neb.com>

Complete QC (Qubit) of pooled libraries.

Pool can be stored at -20C at this stage.

## 7. Adapter Ligation

Perform this step immediately prior to loading the sequencer.  
Ensure the library pool is diluted to loading concentration.

Consider performing a pore scan prior to adapter ligation to ensure the flow cell is okay to load.  
(It's okay to have adapter-ligated library sit on ice for ~30 min to an hour).

**Dilute Adapter (in a fresh 1.5ml Eppendorf DNA LoBind tube):**

|                       |     |
|-----------------------|-----|
| Adapter Buffer (ADB)  | 3.5 |
| Rapid Adapter (RA)    | 1.5 |
| Total diluted adapter | 5   |

**Add diluted adapter to the 11  $\mu$ L barcoded DNA**

|                            |    |
|----------------------------|----|
| Volume of pool             | 11 |
| Diluted adapter ( $\mu$ L) | 1  |

Mix gently by pipeting up and down.

Incubate for 5 min at room temperature.

The prepared library is ready to be loaded into the MinION flow cell. Store the library on ice until ready to load.

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## 8. Load the Sequencer

Load the sequencer according to the ONT MinION Flowcell Priming and Loading protocol

## MODULE 2: DRY LAB

*Analysis and Interpretation of Sequencing Data  
Utilizing CZ ID and Other Bioinformatics Tools*

**[WEEK 2 DATES]**

# CZ ID TRAINING: ONT REFERENCE TABLES

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QC Metrics Table - ONT

| QC Metric                   | Definition                                                                                                                                                                                                 |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Passed Filters</b>       | The percentage of reads remaining after host filtering, quality filtering and subsampling, compared to the initial reads input. Tells you how many reads were used for final analysis.                     |
| <b>Passed QC</b>            | Percentage of reads remaining after QC filtering to remove low quality bases (phred score < 9), short reads (< 100 bp), and low complexity reads.                                                          |
| <b>Total Reads*</b>         | The total number of reads uploaded.                                                                                                                                                                        |
| <b>Subsampled Fraction*</b> | After QC filtering and host/human data removal, the remaining reads are subsampled to 1 million. This fraction specifies the ratio of subsampled reads to total reads passing QC and host/human filtering. |

*\* Fields are not default in the samples table and need to be added by clicking on the column selector*

*The Sample Table only provides stats at the read level for you to keep track of how many reads made it through data pre-processing steps (e.g., QC filtering and subsampling) and, thus, were used for microbe detection. If you are interested in information at the base pair level, please see the Sample Overview report available through [bulk download](#) or look at the pipeline details on the Sample Report page.*

Sample Metrics Table

| Report Metric           | Definition                                                                                                                                                                                                                                                                                                     |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bases per million (bPM) | <p>Number of bases within all the reads aligning to a given taxon, including those assembled into contigs that mapped to the taxon, per million bases sequenced</p> $\text{bPM} = \left( \frac{\text{Bases matching taxon}}{\text{Total Bases} \times \text{Subsampled Fraction}} \right) (1 \text{ Million})$ |
| Bases (b)               | <p>Number of bases within all the reads aligning to a given taxon, including those assembled into contigs that mapped to the taxon</p>                                                                                                                                                                         |
| Reads (r)               | <p>Number of reads aligning to a given taxon in the NCBI NT/NR database, including those assembled into contigs that mapped to the taxon</p>                                                                                                                                                                   |

|                         |                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Contigs (contig)        | Number of contigs aligning to the taxon in the NCBI NT/NR database                                                                                                                                                                                                                                                                                                                                |
| Contig bases (contig b) | Number of bases within all the reads that assembled into contigs aligning to a given taxon                                                                                                                                                                                                                                                                                                        |
| Percent identity (%id)  | Average percent identity between all the query sequences (contigs and unassembled reads) and their matching taxon                                                                                                                                                                                                                                                                                 |
| Length (L)              | Average length of alignments between all the query sequences (contigs and unassembled reads) and their matching taxon. Note that values against NR are reported in base pairs                                                                                                                                                                                                                     |
| Expect value (E value)  | Average expect value (E-value) of alignments against the NT and NR databases. The E-value represents the number of matches with similar quality one would “expect” to see by random chance. This parameter provides a measure of randomness. The lower the E-value, the lower the probability of getting a match or alignment by random chance (i.e., the closer the E-value is to 0, the better) |

## Pipeline Table

*Metrics are accessible through Sample Details -> Pipelines -> Bases remaining*

| Pipeline Step | Function                                                                                                                                                                                  | If Many Reads Lost During this Step...                                                                                                                                                                                                                                                         |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| fastp         | Low quality read removal<br><br>Low complexity read removal<br><br>Note: adapter trimming is <i>not</i> completed in CZID. It must be done in ONT MinKNOW/Dorado, prior to upload to CZID | Many reads of low-quality (Phred scores, GC content); check sequencing run - suggest reviewing sequencing run report<br><br>Reads are relatively low-complexity, containing homopolymer nucleotides or simple sequence repeats<br><br>Many short reads: possible degradation, low quality RNA. |
| minimap2      | Host and human filtering                                                                                                                                                                  | Large proportion of reads mapped to the host/human reference genome; this will vary by sample type and sample pre-processing                                                                                                                                                                   |
| seqtk         | Subsampling to reduce computational burden                                                                                                                                                | Sample received a lot of reads during sequencing and/or has a high proportion of reads that are microbial                                                                                                                                                                                      |

# CZ ID TRAINING: WORKSHEETS

## Goal For Your Project

*Why is your team using metagenomics and CZ ID? What problem(s) are you trying to solve with mNGS and how does it relate to your broader research?*

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## QC Worksheet

### QC Step 1: Identify samples with low total reads

*The total number of reads uploaded (after input validation).*

| Metric      | Criterion | How to Interpret                                                                                                                                                                          | Sample Name(s) | Next Steps/Notes |
|-------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------------|
| Total Reads | Outliers  | You should make note of outliers and high variability in Total Reads. This can indicate uneven pooling in the library preparation steps. You may choose to resequence low-output samples. |                |                  |

### QC Step 2: Identify samples with low-quality reads (low passed QC)

*The percentage of reads that passed the quality filtering thresholds imposed by fastp. Gives you an idea of the sequencing quality.*

| Metric    | Criterion | How to Interpret                                                                                                                                                                                                                                                                                                                                                                                                        | Sample Name(s) | Next Steps/Notes |
|-----------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------------|
| Passed QC | >50%      | You can be relatively confident in the quality of the sequencing data.                                                                                                                                                                                                                                                                                                                                                  |                |                  |
|           | <10%      | Samples with low % Passed QC likely contained many short sequences that were associated with adapter dimers or sequences that obtained low-quality scores during base-calling. If the Passed Filters value is below 20%, you should work to understand why reads were filtered out (see next section) and use this information to improve library preparation steps. Interpret results from these samples with caution. |                |                  |
|           | *Outliers | Take note of any values above that stand apart from the others because of very high or very low Passed QC values. These should be noted for their unusually high or low sequencing quality and looked into more closely.                                                                                                                                                                                                |                |                  |

### QC Step 3: Identify samples with low number of reads passing CZ ID filters

*The percentage of reads remaining after host filtering, quality filtering and subsampling, compared to the initial reads input. Tells you how many reads were used for final analysis. Expected reads passing CZ ID filters varies by sample type.*

| Metric         | Criterion | How to Interpret                                                                                                                          | Sample Name(s) | Next Steps/Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Passed Filters | >5%       | A higher Passed Filters value is generally positive as it indicates that you have a greater amount of reads to analyze for non-host taxa. |                | What is the Sample Type for each of the samples? <ul style="list-style-type: none"><li>• Samples with a known microbiome (ie Stool, NP Swab, Environmental Samples, Sputum) tend to have a higher number of non-host reads due to greater microbial content in the samples.</li><li>• Sterile site fluids (Serum, CSF, Plasma) tend to have fewer non-host reads and greater numbers of host reads.</li><li>• Whole Blood tends to have a distribution of non-host reads in between these two extremes due to the high abundance of host cells.</li></ul> |
|                | <5%       | It is not uncommon to see Passed Filters values below 1% for samples with high abundance of host reads.                                   |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                | *Outliers | Take note of any values that stand apart from the others because of very high or very low Passed Filters values.                          |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

#### QC Step 4: Identify at which step most bases are lost during pre-processing

*Look at bases remaining for each sample, accessible through Sample Details -> Pipelines -> Bases remaining*

| Sample | Total bases<br>(Validate input) | Which step were most bases lost? What<br>% of bases were lost at this step? | Sample type | Notes about Bases<br>Remaining metrics | Flag because of<br>QC?   |
|--------|---------------------------------|-----------------------------------------------------------------------------|-------------|----------------------------------------|--------------------------|
|        |                                 |                                                                             |             |                                        | <input type="checkbox"/> |
|        |                                 |                                                                             |             |                                        | <input type="checkbox"/> |
|        |                                 |                                                                             |             |                                        | <input type="checkbox"/> |
|        |                                 |                                                                             |             |                                        | <input type="checkbox"/> |

Sample QC Summary Table

**QC Step 5: Summarize samples with poor QC metric(s) to flag for analysis interpretation**

*Using the bases remaining and QC Steps 1-4 above, fill out the following for low quality samples:*

| Sample Name | QC Flag?                 | Flagged because of which step / metric? | Conclusions |
|-------------|--------------------------|-----------------------------------------|-------------|
|             | <input type="checkbox"/> |                                         |             |
|             | <input type="checkbox"/> |                                         |             |
|             | <input type="checkbox"/> |                                         |             |
|             | <input type="checkbox"/> |                                         |             |

## Sample Report

Think about the following questions when looking at your sample report:

1. What is the goal of your study?
2. Are there any pathogens present that could be causing disease in your set of samples?
  - a. Is it a real hit? Why do you think it is?
3. What are some of the other microbes?
  - a. Why do you think they are there?
4. Did you find any false positives?
  - a. How did you determine this?

## **Analysis Workflow - ONT RNA mNGS for respiratory viruses**

### **Navigate**

1. Go to each sample report
2. Filter category for viruses + non-phage

### **Complete a first pass sample report view**

1. Look at the list of detected taxa
2. Record NT bPM, NT r, and any other relevant metrics
3. Look at the coverage plot for each taxa and record your interpretation (low, medium, high coverage)
4. BLASTn contigs from each candidate taxa and record what you find (specific, non-specific, etc)

### **Check negative water controls**

1. Navigate to the sample report for each negative water control included in the run
2. For each taxa observed in the samples, record the NT bPM, NT r, and any other metrics in each water control
3. If you have >1 water control, calculate the average signal for each taxa

### **Put it all together (background subtraction + interpretation)**

1. Subtract the background/water control signal from the sample signal for each taxa
2. Interpret: real signal should be higher in samples than in water controls

Sample Summary Table (#1)

| Name of sample |                                                                              |                          |                                        |                              |             |                                           |                     |
|----------------|------------------------------------------------------------------------------|--------------------------|----------------------------------------|------------------------------|-------------|-------------------------------------------|---------------------|
| Taxon Name     | Important Metrics (i.e. high bPM, known pathogen, number of contigs, length) | Pathogenic?              | Does it answer your research question? | Coverage Visualization Notes | BLAST Notes | Why do you think this microbe is present? | Is this a real hit? |
|                |                                                                              | <input type="checkbox"/> |                                        |                              |             |                                           |                     |
|                |                                                                              | <input type="checkbox"/> |                                        |                              |             |                                           |                     |
|                |                                                                              | <input type="checkbox"/> |                                        |                              |             |                                           |                     |
|                |                                                                              | <input type="checkbox"/> |                                        |                              |             |                                           |                     |

Sample Summary Table (#2)

| Name of sample |                                                                              |                          |                                        |                              |             |                                           |                     |
|----------------|------------------------------------------------------------------------------|--------------------------|----------------------------------------|------------------------------|-------------|-------------------------------------------|---------------------|
| Taxon Name     | Important Metrics (i.e. high bPM, known pathogen, number of contigs, length) | Pathogenic?              | Does it answer your research question? | Coverage Visualization Notes | BLAST Notes | Why do you think this microbe is present? | Is this a real hit? |
|                |                                                                              | <input type="checkbox"/> |                                        |                              |             |                                           |                     |
|                |                                                                              | <input type="checkbox"/> |                                        |                              |             |                                           |                     |
|                |                                                                              | <input type="checkbox"/> |                                        |                              |             |                                           |                     |
|                |                                                                              | <input type="checkbox"/> |                                        |                              |             |                                           |                     |

Sample Summary Table (#3)

Name of sample

| Taxon Name | Important Metrics (i.e. high bPM, known pathogen, number of contigs, length) | Pathogenic?              | Does it answer your research question? | Coverage Visualization Notes | BLAST Notes | Why do you think this microbe is present? | Is this a real hit? |
|------------|------------------------------------------------------------------------------|--------------------------|----------------------------------------|------------------------------|-------------|-------------------------------------------|---------------------|
|            |                                                                              | <input type="checkbox"/> |                                        |                              |             |                                           |                     |
|            |                                                                              | <input type="checkbox"/> |                                        |                              |             |                                           |                     |
|            |                                                                              | <input type="checkbox"/> |                                        |                              |             |                                           |                     |
|            |                                                                              | <input type="checkbox"/> |                                        |                              |             |                                           |                     |

Sample Summary Table (#4)

Name of sample

| Taxon Name | Important Metrics (i.e. high bPM, known pathogen, number of contigs, length) | Pathogenic?              | Does it answer your research question? | Coverage Visualization Notes | BLAST Notes | Why do you think this microbe is present? | Is this a real hit? |
|------------|------------------------------------------------------------------------------|--------------------------|----------------------------------------|------------------------------|-------------|-------------------------------------------|---------------------|
|            |                                                                              | <input type="checkbox"/> |                                        |                              |             |                                           |                     |
|            |                                                                              | <input type="checkbox"/> |                                        |                              |             |                                           |                     |
|            |                                                                              | <input type="checkbox"/> |                                        |                              |             |                                           |                     |
|            |                                                                              | <input type="checkbox"/> |                                        |                              |             |                                           |                     |

## Summary list of candidate taxa

Look at the Sample Summary Tables above and the sample report for each sample.

Suggested starting threshold filters:

*NT bPM  $\geq 100$ ; NR bPM  $\geq 1$  (only if there are contigs); NT L  $> 75$ ; NT E-value  $< 0.001$  ( $10^{-3}$ ); NR E-value  $< 0.001$  ( $10^{-3}$ )*

|        | CANDIDATE TAXA (what do you think is in your sample?) |                        |                                             |
|--------|-------------------------------------------------------|------------------------|---------------------------------------------|
| Sample | Initial Sample Report review<br>(no filters)          | With threshold filters | With threshold filters +<br>category filter |
|        |                                                       |                        |                                             |
|        |                                                       |                        |                                             |
|        |                                                       |                        |                                             |
|        |                                                       |                        |                                             |

1. Are any of the candidate organisms present in the water samples?
  - a. If so, are they present in any other samples?
2. Which organisms appear consistently across all samples, but not in water control?
3. If you have metadata, are there microbes present in one set of samples that aren't in another (i.e. cases vs controls)?

Identification of Background Signal

For each negative water or extraction or negative sample control, collect metrics about the candidate taxa identified in the previous step.

|                                             |            | SIGNAL IN WATER CONTROLS for CANDIDATE TAXA (account for background) |                                          |                             |                           |
|---------------------------------------------|------------|----------------------------------------------------------------------|------------------------------------------|-----------------------------|---------------------------|
| Candidate taxa<br><i>From previous step</i> | Neg Ctrl   | NT bpm<br><i>for each negative control</i>                           | NT r<br><i>for each negative control</i> | NT bpm average in neg ctrls | NT r average in neg ctrls |
| betacoronavirus                             | Water 1    | 241                                                                  | 1                                        | 292.7                       | 2.33                      |
|                                             | Water 2    | 253                                                                  | 1                                        |                             |                           |
|                                             | Neg Ctrl 3 | 384                                                                  | 5                                        |                             |                           |
|                                             |            |                                                                      |                                          |                             |                           |
|                                             |            |                                                                      |                                          |                             |                           |

|                                                    |                   | SIGNAL IN WATER CONTROLS for CANDIDATE TAXA (account for background) |                                                 |                                    |                                  |
|----------------------------------------------------|-------------------|----------------------------------------------------------------------|-------------------------------------------------|------------------------------------|----------------------------------|
| <b>Candidate taxa</b><br><i>From previous step</i> | <b>Neg Ctrl</b>   | <b>NT bPM</b><br><i>for each negative control</i>                    | <b>NT r</b><br><i>for each negative control</i> | <b>NT bPM average in neg ctrls</b> | <b>NT r average in neg ctrls</b> |
| <i>betacoronavirus</i>                             | <i>Water 1</i>    | <i>241</i>                                                           | <i>1</i>                                        | <i>292.7</i>                       | <i>2.33</i>                      |
|                                                    | <i>Water 2</i>    | <i>253</i>                                                           | <i>1</i>                                        |                                    |                                  |
|                                                    | <i>Neg Ctrl 3</i> | <i>384</i>                                                           | <i>5</i>                                        |                                    |                                  |
|                                                    |                   |                                                                      |                                                 |                                    |                                  |

|                                                    |                 |                                                                                         |                                                 |                                    |                                  |
|----------------------------------------------------|-----------------|-----------------------------------------------------------------------------------------|-------------------------------------------------|------------------------------------|----------------------------------|
|                                                    |                 | <b>(continued) SIGNAL IN WATER CONTROLS for CANDIDATE TAXA (account for background)</b> |                                                 |                                    |                                  |
| <b>Candidate taxa</b><br><i>From previous step</i> | <b>Neg Ctrl</b> | <b>NT bPM</b><br><i>for each negative control</i>                                       | <b>NT r</b><br><i>for each negative control</i> | <b>NT bPM average in neg ctrls</b> | <b>NT r average in neg ctrls</b> |
|                                                    |                 |                                                                                         |                                                 |                                    |                                  |
|                                                    |                 |                                                                                         |                                                 |                                    |                                  |
|                                                    |                 |                                                                                         |                                                 |                                    |                                  |
|                                                    |                 |                                                                                         |                                                 |                                    |                                  |
|                                                    |                 |                                                                                         |                                                 |                                    |                                  |
|                                                    |                 |                                                                                         |                                                 |                                    |                                  |

**Background Subtraction**

*Put it all together! Subtract average background signal (previous step) from signal found in samples (two steps before this).  
A real hit will be found at higher levels in the sample than in the negative controls*

|        |                   | SIGNAL IN WATER CONTROLS and SAMPLES for<br>CANDIDATE TAXA |                 |                                   |                                 |                                          |                                           |
|--------|-------------------|------------------------------------------------------------|-----------------|-----------------------------------|---------------------------------|------------------------------------------|-------------------------------------------|
| Sample | Candidate<br>taxa | NT bPM:<br>sample                                          | NT r:<br>sample | NT bPM:<br>average in<br>neg ctrl | NT r:<br>average in<br>neg ctrl | NT bPM<br>(sample - neg<br>ctrl average) | NT r<br>(sample -<br>neg ctrl<br>average) |
|        |                   |                                                            |                 |                                   |                                 |                                          |                                           |
|        |                   |                                                            |                 |                                   |                                 |                                          |                                           |
|        |                   |                                                            |                 |                                   |                                 |                                          |                                           |
|        |                   |                                                            |                 |                                   |                                 |                                          |                                           |
|        |                   |                                                            |                 |                                   |                                 |                                          |                                           |

## Potential Hit List

*Record the hits for each sample that pass all filter steps and have positive signal above the negative control background.*

*This is your potential hit list!*

*Because mNGS sequencing is a research tool, we recommend completing some of the following steps to increase confidence in the hit.*

- 1. Reprep and sequence sample starting from a fresh extraction*
- 2. Resequence the completed library individually or in a pool without other samples with high abundance of the taxon (reduce barcode bleeding)*
- 3. Perform other confirmatory tests on the sample like qPCR*
- 4. Consider more advanced bioinformatics analysis to compare SNP pattern across samples*

|        | POTENTIAL HIT LIST                          |       |            |
|--------|---------------------------------------------|-------|------------|
| Sample | Taxa detected above thresholds + background | Notes | Next steps |
|        |                                             |       |            |
|        |                                             |       |            |
|        |                                             |       |            |

# CZ ID TRAINING: ILLUMINA REFERENCE TABLES

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QC Metrics Table

| QC Metric                   | Definition                                                                                                                                                                                                                                                    |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Passed Filters</b>       | The percentage of reads remaining after host filtering, quality filtering and subsampling, compared to the initial reads input. Tells you how many reads were used for final analysis.                                                                        |
| <b>Passed QC</b>            | The percentage of reads that passed the quality filtering thresholds imposed by fastp. Gives you an idea of the sequencing quality.                                                                                                                           |
| <b>Total Reads*</b>         | The total number of reads uploaded.                                                                                                                                                                                                                           |
| <b>DCR*</b>                 | Duplicate Compression Ratio; The ratio of sequences present prior to running CZID dedup (duplicate removal) vs after duplicate removal.                                                                                                                       |
| <b>ERCC reads*</b>          | The total number of reads aligning to <a href="#">ERCC</a> (External RNA Controls Consortium) sequences. The number of ERCC reads will be proportionally higher in samples with relatively low input RNA as compared to those with high amounts of input RNA. |
| <b>SubSampled Fraction*</b> | After host filtration and QC, the remaining reads are subsampled to 1 million fragments (2 million paired reads). This field indicates the ratio of subsampled reads to total reads passing host filtration and QC steps.                                     |

\* Fields are not default in the samples table and need to be added by clicking on the column selector

**Sample Metrics Table**

| Report Metric | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Score         | Score is only calculated if a background model is applied. It is CZ ID's heuristic for ranking microbial hits. The score is intended to combine the following aspects of the evidence for a hit: (a) species-level information, (b) genus-level information, (c) information about relative abundance within the sample, (d) information about abundance relative to the chosen background controls. The score is calculated as follows: $((\text{abs}(\text{genus NT Z}) * \text{species NT Z} * \text{species NT rPM}) + (\text{abs}(\text{genus NR Z}) * \text{species NR Z} * \text{species NR rPM}))$ |
| Z             | Z-score statistic is only computed if a background model is applied. It is used for evaluating the prevalence of microbes in your sample as compared to background controls. The Z-score is computed based on the specified background model.                                                                                                                                                                                                                                                                                                                                                              |
| rPM           | Number of reads aligning to the taxon in the NCBI NT/NR database, per million reads sequenced                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| r             | Number of reads aligning to the taxon in the NCBI NT/NR database                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| contig        | Number of assembled contigs aligning to the taxon in the NCBI NT/NR database                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|             |                                                                                                                                                                                                                                                       |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| contig<br>r | Total number of reads aligning to all assembled contigs for this taxon                                                                                                                                                                                |
| %id         | Average percent-identity of alignments to NCBI NT/NR                                                                                                                                                                                                  |
| L           | Average length of the local alignment for all contigs and reads assigned to this taxon                                                                                                                                                                |
| E<br>value  | Average Expect value (e-value) of alignments to NCBI NT/NR. The Expect value (e-value) is a parameter that describes the number of hits one can "expect" to see by chance when searching a database of a particular size. The closer to 0 the better. |

**Pipeline Table**

| <b>Pipeline Step</b>                                 | <b>Function</b>                       | <b>If Many Reads Lost During this Step...</b>                                                       |
|------------------------------------------------------|---------------------------------------|-----------------------------------------------------------------------------------------------------|
| fastp                                                | Adapter Trimming                      | Many short reads, possible adapter dimers, possible degradation, possible over-fragmentation.       |
|                                                      | Low quality read removal              | Many reads of low-quality (Phred scores, GC content, over-clustering).                              |
|                                                      | Low complexity read removal           | Reads are relatively low-complexity - containing homopolymer nucleotides or simple sequence repeats |
| HISAT2, Bowtie 2                                     | Host Subtraction                      | Large proportion of reads mapped to the host reference genome                                       |
| HISAT2, Bowtie 2                                     | Remove human reads regardless of host | Large proportion of reads mapped to human reference genome                                          |
| CZID-dedup<br>(*Available through "reads remaining") | Remove duplicates                     | Many duplicate reads, possible over-amplification during library preparation.                       |



# APPENDIX

## A1. Optional Secondary PCR Amplification

| Reagent                              | 1 rxn          | X full rxns                                     |
|--------------------------------------|----------------|-------------------------------------------------|
|                                      |                | X =                                             |
|                                      | vol stock (μL) | vol stock (μL)                                  |
| Clean barcoded DNA                   | 9              | 9 each                                          |
| Nuclease-free water                  | 15.5           |                                                 |
| LongAmp Taq 2x MM                    | 25             |                                                 |
| RLB (01-24, at 10 μM)*               | 0.5            | 0.5 μL unique to each sample                    |
| Master Mix volume                    | 40.5           |                                                 |
| <b>Total volume</b>                  | <b>50</b>      | <b>50 each</b>                                  |
| Pink = Reagents to add to master mix |                | Add 40.5 μL of MM and 0.5 μL RLB to tagged cDNA |

\*It is critical to use the same barcodes as initial PCR amplification step

\*\*Add barcodes last

Mix MM and final rxn. by pipeting up and down and start the following thermocycling program:

| Thermocycler (lid at 105°C) | Cycles |
|-----------------------------|--------|
| 95 °C for 45s               | 1      |
| 95 °C for 15s               | 5-10   |
| 59 °C for 15s               |        |
| 65 °C for 5 min             |        |
| 65°C for 10 mins            | 1      |
| Hold at 10°C                |        |

After completion of PCR, repeat Steps 4 and 5 (SPRI clean-up and QC of individual libraries). If libraries are still too low concentration at this stage, proceed with pooling and loading.

## A2. DNase I treatment - ZYMO

The protocols included in this booklet are outlined for **RNA** metagenomic library prep. Generally, DNase I treatment of input nucleic acid is recommended to isolate RNA and reduce host/human signal from DNA. It may be completed on-column during the extraction or separately after the extraction. The need for DNase I treatment may vary depending on the extraction kit.

| Reagent                                   | 50 $\mu$ L sample    | 20 $\mu$ L sample    | X 20 $\mu$ L rxns               |
|-------------------------------------------|----------------------|----------------------|---------------------------------|
|                                           |                      |                      | X =                             |
|                                           | vol stock ( $\mu$ L) | vol stock ( $\mu$ L) | vol stock ( $\mu$ L)            |
| TNA sample                                | 40                   | 20                   | 20 each                         |
| DNA Digestion Buffer (Zymo)               | 5                    | 2.50                 |                                 |
| DNase I stock (1U/ $\mu$ L) (Zymo, E1010) | 5                    | 2.50                 |                                 |
| Water                                     | -                    |                      |                                 |
| Master Mix volume                         | 10                   | 5.00                 |                                 |
| <b>Total volume</b>                       | <b>50</b>            | <b>25.00</b>         | <b>25 each</b>                  |
| Green = Reagents to add to master mix     |                      |                      | Add 5 $\mu$ L mix to TNA sample |

Incubate for 15min at RT  
Do a SPRI clean-up as follows

### SPRI Bead Clean-up

|                                   |       |         |
|-----------------------------------|-------|---------|
| SPRI bead ratio:                  | 2     |         |
| SPRI to add (per half rxn):       | 50.00 | $\mu$ L |
| Total SPRI bead volume:           | 50.00 | $\mu$ L |
| Elution volume (per half rxn):    | 17    | $\mu$ L |
| Aspiration volume (per half rxn): | 15    | $\mu$ L |

For SPRI protocol, refer to Page 5.

### A3. Nanopore RNA Metagenomics Diagnostic PCR

**Project:**

**Date:**

Goal: obtain estimated library size for library QC

If you are unable to use a high sensitivity capillary electrophoresis system (like TapeStation), you can perform a diagnostic PCR instead as a proxy for the overall library size. Do note that this method tends to slightly overestimate the size of the library.

**Primer for use in Diagnostic PCR:**

Front adapter portion of barcoding PCR primer (RLB-PCR from SMART-9N publication):

5' - TTTTCGTGCGCCGCTTCA - 3'

#### Appendix 1. Diagnostic PCR for Library QC

| Reagent                                | 1 rxn          | X full rxns*                                         |
|----------------------------------------|----------------|------------------------------------------------------|
|                                        | vol stock (μL) | X = <span style="background-color: yellow;"> </span> |
| Clean barcoded DNA                     | 1              |                                                      |
| Nuclease-free water                    | 23.5           |                                                      |
| LongAmp Taq 2x MM                      | 25             |                                                      |
| RLB-PCR Primer (10μM)                  | 0.5            |                                                      |
| Master Mix volume                      | 48.5           |                                                      |
| <b>Total volume</b>                    | <b>50</b>      |                                                      |
| Orange = Reagents to add to master mix |                | Add 49 μL of MM to 1 μL of library                   |

\* Half reactions should be okay to run for the diagnostic PCR

Mix MM and final rxn. by pipeting up and down and start the following thermocycling program:

| Thermocycler (lid at 105°C) | Cycles |
|-----------------------------|--------|
| 95 °C for 45s               | 1      |
| 95 °C for 15s               | 20     |
| 56 °C* for 15s              |        |
| 65 °C for 5 min             |        |
| 65°C for 10 mins            | 1      |
| Hold at 10°C                |        |

#### Run Gel Electrophoresis

Run PCR amplified product on a 1% gel with 1kb ladder to visualize. The volumes used will depend on the gel electrophoresis system being used.